MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCFS200 - DATA STRUCTURE USING C++

(2024 ADMISSION ONWARDS)

QPCODE: 25901201

ANSWER SCHEME

Part A

Answer any 10 questions

Each question carries 1 marks

1. Describe the "Traversing" operation as it applies to data structures.

Traversing is the process of visiting each element/node of a data structure exactly once to access or process the data stored in it.

2. State the meaning of subscript.

A subscript is an index or position number used to identify an element in an array.

3. Differentiate between insertion at the beginning and at the end of a linked list.

Beginning insertion adds a new node before the first node and updates the head pointer. End insertion adds a new node after the last node and updates the last node link.

4. What is a linkedlist?

A linked list is a linear data structure consisting of nodes where each node contains data and a pointer/reference to the next node.

5. In an input-restricted queue, from which end can insertion occur?

Insertion can occur only from one end, usually the rear end, while deletion can occur from both ends.

6. Examine the relationship between parentheses and stack usage in arithmetic expressions.

Parentheses are handled using stacks. Opening parentheses are pushed and closing parentheses cause popping to check matching pairs.

7. Identify the principle followed by a stack.

Stack follows the LIFO (Last In First Out) principle.

8. Compare the roles of push and pop operations in maintaining stack integrity.

Push inserts an element into the stack, while pop removes the top element. Together they maintain proper stack order.

9. Name the variant of a linked list in which none of the node contains NULL pointer.

Circular linked list.

10. List the data structures, other than arrays, where linear search can be effectively applied.

Linked lists, stacks, queues etc..

11. Outline the steps involved in preorder traversal of a binary tree.

Visit root, traverse left subtree, then traverse right subtree (Root-Left-Right).

12. Predict the index of left child of a node at index 4.

For array index starting at 0, left child index = 2i + 1. For i = 4, left child = 9.

For array index starting at 1, 2i= 2*4= 8

Part B

Answer any 4 questions

Each question carries 5 marks

13. Name and briefly describe three common operations performed on linear data structures.

1. Insertion - Adding a new element.

2. Deletion - Removing an existing element.

3. Traversal - Visiting elements one by one.

14. Explain priority queue with memory representation.

A priority queue is a queue where each element has a priority. The element with highest priority is removed first. It can be represented using arrays or linked lists with priority fields. Draw diagram of memory representation

15. Give a sequence of stack operations analyze and show the status of the stack after each step.

Example:

Push(10) → Stack: 10

Push(20) → Stack: 10,20

Push(30) → Stack: 10,20,30

Pop() → removes 30

Stack: 10,20 Give any example like the above one

16. Explain the rule for pushing an operator onto the stack when the stack already contains an operator of equal or higher precedence.

If the stack top has an operator of equal or higher precedence, the existing operator is popped before pushing the new operator.

17. Differentiate between strict binary tree and height balanced binary tree.

Strict binary tree: Every node has either zero or two children.

Height balanced binary tree: Difference between heights of left and right subtrees is limited.

18. Differentiate between complete binary tree and strict binary tree.

Complete binary tree: All levels are completely filled except possibly the last, which is filled from left.

Strict binary tree: Every node has either 0 or 2 children.

Part C

Answer any 2 questions

Each question carries 10 marks

19. Describe the representation of a two-dimensional array in memory. Explain how elements are stored linearly even though they are logically arranged in rows and columns.

A two-dimensional array is stored in contiguous memory locations. In row-major order, all elements of the first row are stored followed by the next rows. In column-major order, elements of columns are stored consecutively. Address calculation is used to access elements. Give explanation.

20. Describe the various steps to evaluate a postfix expression with an example.

Steps:

1. Scan postfix expression from left to right.

2. If operand is found, push it into stack.

3. If operator is found, pop required operands.

4. Perform operation and push result back.

5. Final stack value gives the answer.

Give any example

21. Explain different types of trees with examples.

Types of trees:

1. Binary Tree

2. Strict Binary Tree

3. Complete Binary Tree

4. Binary Search Tree

5. AVL Tree etc..

Give example and explanation